

Unit 5, Lesson 4: Solving Problems Involving Real-World Scale Drawings

Benchmark

MA.7.GR.1.5 Solve mathematical and real-world problems involving dimensions and areas of geometric figures, including scale drawings and scale factors.

Learning Target

- ☐ I can solve real-world problems involving scale drawings.

5.4.1 Warm-Up: A Day in the Life

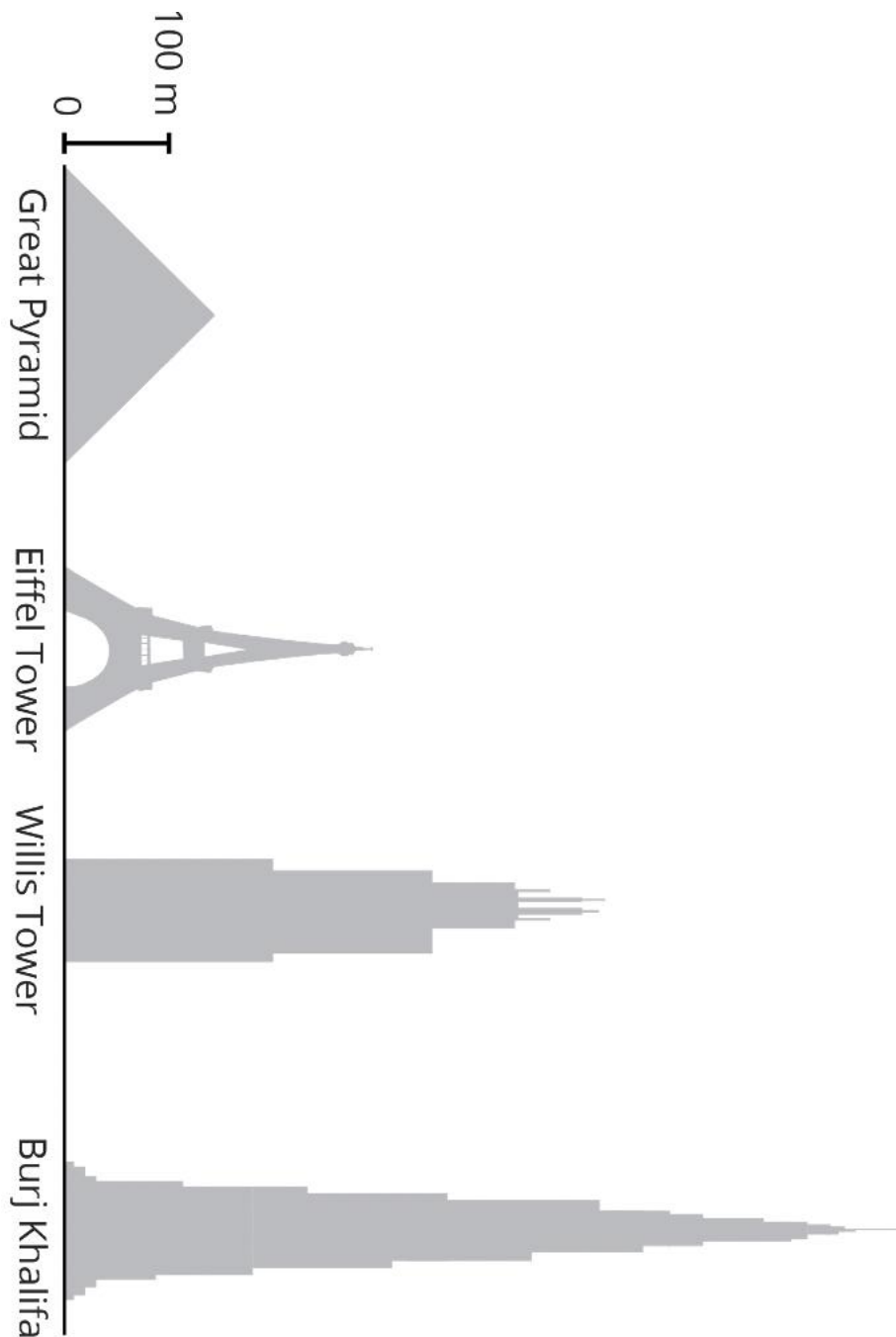
Rachele Louis is an architectural designer who specializes in designing the cores and shells of buildings, or as she calls it, the “skin and bones.” Geometry is key to her work, especially in the initial design phases and when she’s trying to connect forms. She states that there is an art to architecture as well. The appearance of a building defines people’s understanding of the space and evokes certain feelings.

Rachele shares that this integration of art, geometry, and design is her favorite part of the job.

1. Can you think of any examples of how art is used in architecture?
2. What geometric shapes do you see in your classroom?

5.4.2 Guided Instruction: Scale in Architecture

Shown is a scale drawing of some of the world's tallest structures.



1. Turn your paper sideways and measure the line segment that shows the scale with a ruler to the nearest tenth of a centimeter (cm).

2. Express the scale of the drawing using numbers and words.

3. Complete the table using the scaled drawing. Include units.

	Actual Height
Great Pyramid	
Eiffel Tower	
Willis Tower	
Burj Khalifa	

4. About how much taller is the Burj Khalifa than the Eiffel Tower?

5. The tallest mountain in the United States, Mount Denali in Alaska, is about 6,190 meters (m) tall. If this mountain were shown on the scale drawing, explain how its height would compare to the heights of the other structures.

5.4.3 Guided Practice: Accuracy of Models

1. Lego has created a line of products that are scales of important architectural pieces from around the world. Myra saw the Lego statue shown and wondered if it was a scale model of the actual Statue of Liberty.



Lego Statue
height 17 inches (in.)
width 5 in.
depth 5 in.



Actual Statue
height 305 feet (ft)
width 35 ft
depth 35 ft

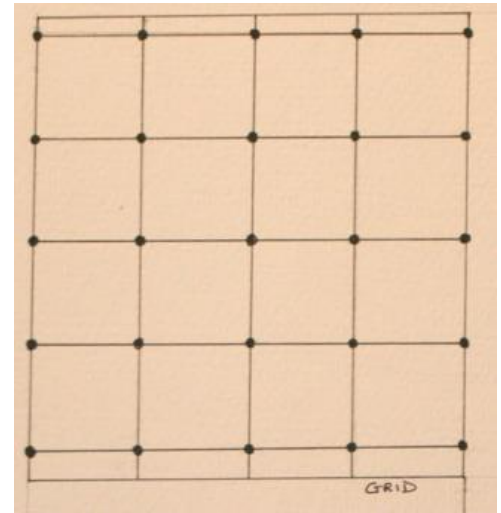
- a. Using the measurements given, determine if the Lego is a scale model of the actual Statue of Liberty.
- b. Explain whether the measurements are enough to determine exactness for the whole scale model.

2. Villa Savoye is a house designed by the Swiss architect Le Corbusier. It was built between 1928 and 1931, and is considered a prime example of modern architecture and the International Style.

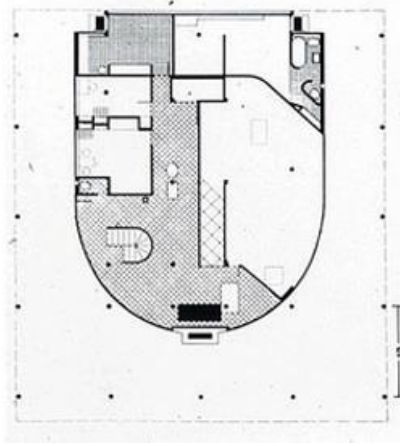
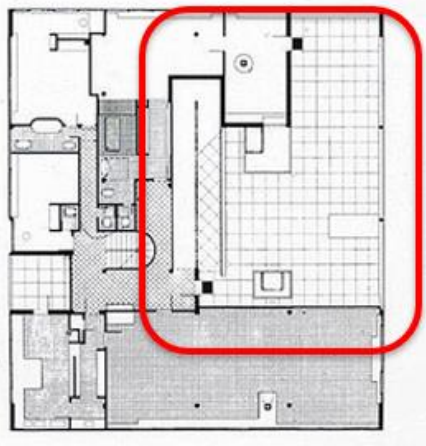
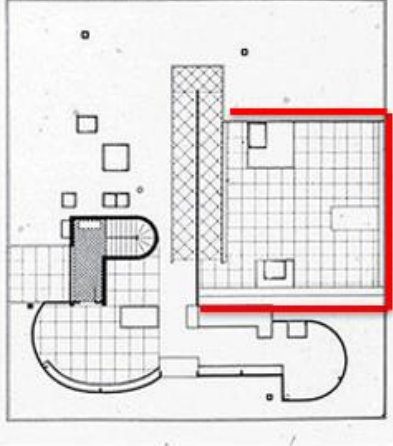
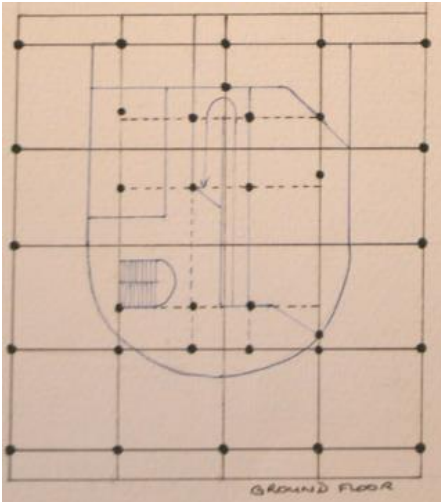
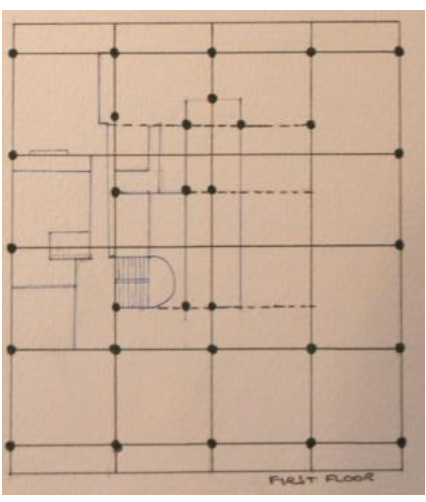
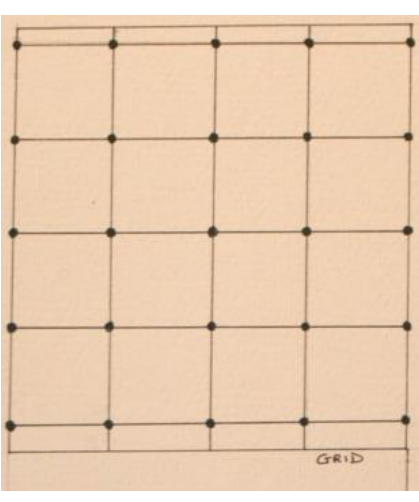
The columns of the structure were placed in an arrangement the architect called a “square grid.” An image of the grid is shown.



- a. If the columns are 4.75 m apart, what is the approximate length and width of the actual house?



The floor plans of Villa Savoye and images of the square grid are shown.

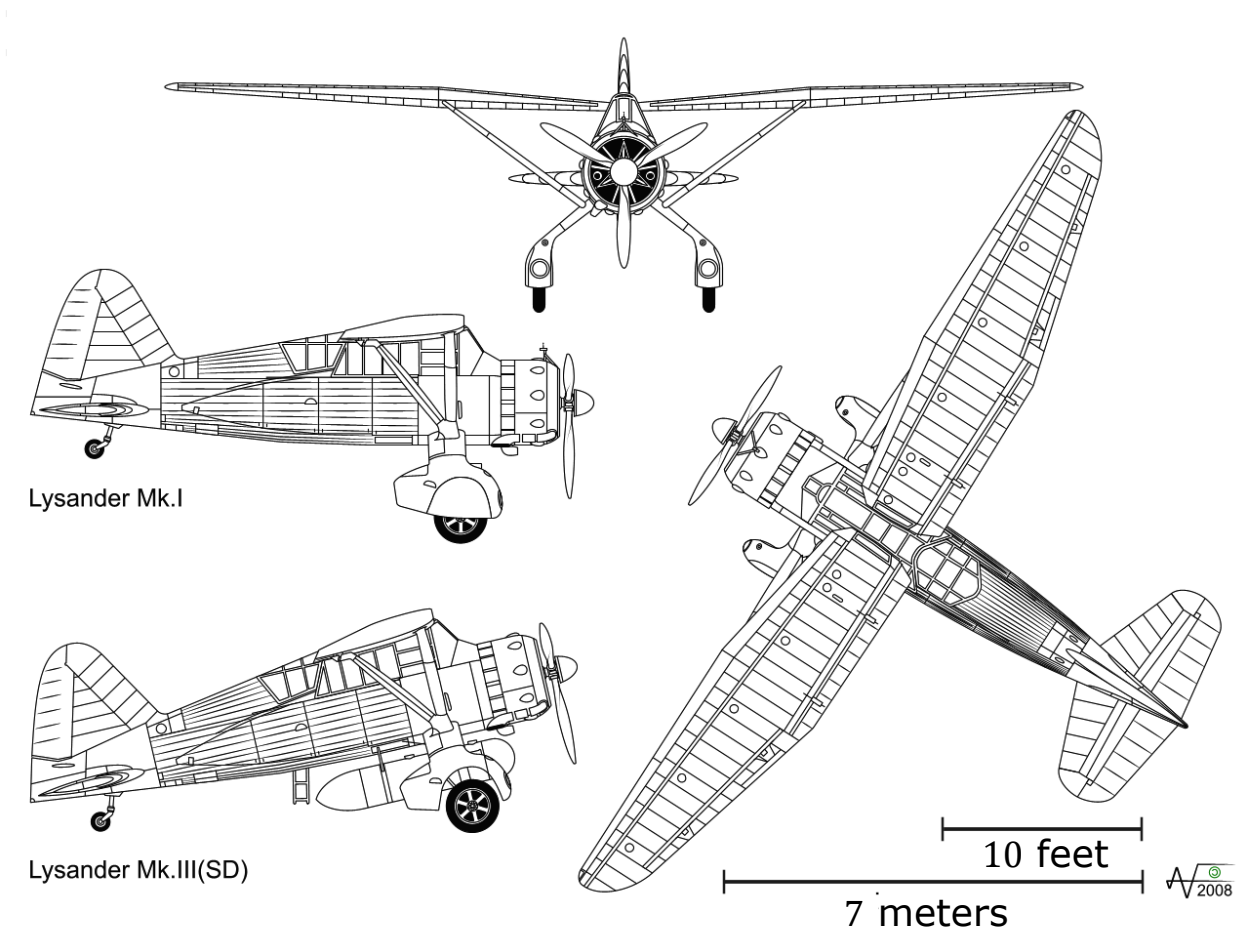
Ground Floor	First Floor	Second Floor
		
		

- b. Approximately how many square meters (m^2) of concrete does the foundation cover?

- c. What is the approximate area of the living room (the tiled area inside the red boxed area on the first floor)?
- d. The plumber needs to put in all new plumbing in one room on the second floor. She only needs to lay the plumbing along three of the walls (the ones lined with red). Calculate approximately how many meters of pipe the plumber needs.

5.4.4 Your Turn!

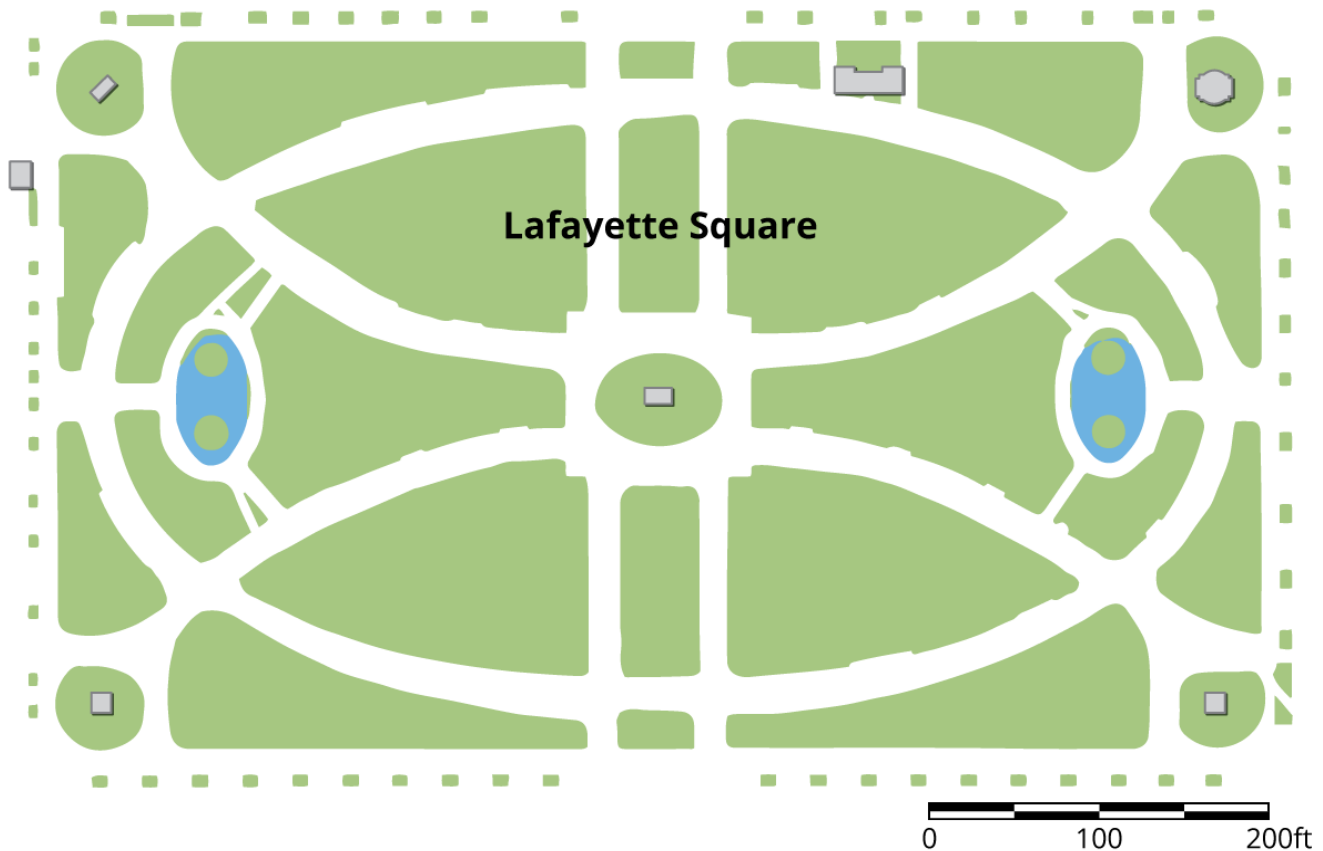
1. The Westland Lysander was an aircraft used by the Royal Air Force in the 1930s. Here are some scale drawings that show the top, side, and front views of the Lysander.



Use the scales and scale drawings to approximate the actual lengths.

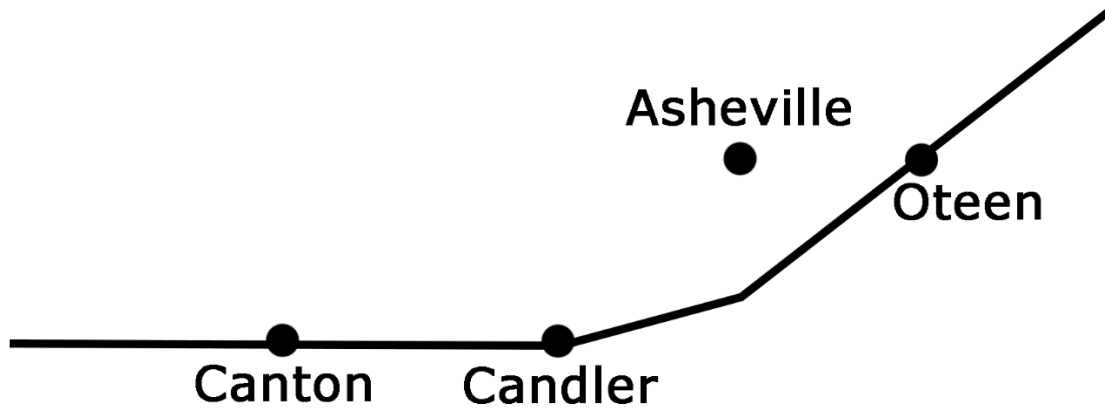
wingspan of the plane to the nearest foot	_____ ft
height of the plane to the nearest foot	_____ ft
length of the plane to the nearest meter	_____ m

2. Below is a scale map of Lafayette Square, a rectangular garden north of the White House.



- a. The scale is shown in the lower right corner. What are the actual side lengths of Lafayette Square in feet?
- b. Use an inch ruler to measure the line segment of the graphic scale. About how many feet does one inch represent on this map?

3. On the map below, $\frac{1}{4}$ in. represents one mile (mi). Candler, Canton, and Oteen are three cities on the map.



- a. If the distance between the actual towns of Candler and Canton is 9 mi., how far apart are Candler and Canton on the map?
- b. If Candler and Oteen are $3\frac{1}{2}$ in. apart on the map, what is the actual distance between Candler and Oteen in miles?

5.4.5 Wrap-Up: Cool Down

1. Write a two-dollar (or more) summary about scale drawings. Each word is worth **10** cents. Be sure to include how scale drawings are used, an example, and how to find measurements from a scale drawing.